

Multi-Camera Stereo Metrology for Cylindrical Objects: A Rigorous Mathematical Analysis of Sub-100 μm Edge Reconstruction

Mathematical Analysis Report

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Abstract

We present a rigorous, first-principles mathematical analysis of multi-camera stereo vision for measuring the edge geometry of a cylindrical object (soda can: diameter 66 mm, height 123 mm) to precisions of 100 μm (1/10 mm) and 10 μm (1/100 mm). We derive the complete epipolar geometry for silhouette-based cylinder measurement, analyze sub-pixel edge localization methods, establish a quantitative error budget, and validate all results through Monte Carlo simulation. We demonstrate that 100 μm precision is achievable with 4 calibrated cameras and standard industrial optics, while 10 μm precision requires ≥ 8 cameras with telecentric or long-focal-length lenses and sub-0.05-pixel edge detection.

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1 Problem Definition and Physical Setup

1.1 Object Model

We model the soda can as a right circular cylinder with:

- Radius: $R = 33$ mm
- Height: $H = 123$ mm
- Axis: aligned with the z -axis, centered at the origin

The cylinder surface is parameterized as:

$$\mathcal{S} = \left\{ \mathbf{P}(\theta, z) = \begin{pmatrix} R \cos \theta \\ R \sin \theta \\ z \end{pmatrix} : \theta \in [0, 2\pi), z \in [-H/2, H/2] \right\} \quad (1)$$

The outward unit normal at any surface point is:

$$\mathbf{n}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix} \quad (2)$$

1.2 Camera Model

Each camera i is modeled as a pinhole camera with intrinsic matrix:

$$\mathbf{K}_i = \begin{pmatrix} f_{x,i} & 0 & c_{x,i} \\ 0 & f_{y,i} & c_{y,i} \\ 0 & 0 & 1 \end{pmatrix} \quad (3)$$

where $f_{x,i} = f_i/p_x$ and $f_{y,i} = f_i/p_y$ are the focal lengths in pixels, f_i is the physical focal length, p_x, p_y are pixel pitches, and $(c_{x,i}, c_{y,i})$ is the principal point.

Reference parameters:

- Sensor: $\frac{1}{2}$ " format, $6.4 \text{ mm} \times 4.8 \text{ mm}$
- Resolution: 2448×2048 pixels
- Pixel pitch: $p = 6.4/2448 \approx 2.614 \mu\text{m}$
- Focal lengths tested: $f \in \{12, 25, 50, 75, 100\} \text{ mm}$
- Working distance: $Z_w \approx 300 \text{ mm}$ (camera to cylinder center)

The extrinsic parameters for camera i consist of rotation $\mathbf{R}_i \in SO(3)$ and translation $\mathbf{t}_i \in \mathbb{R}^3$, giving the projection:

$$\mathbf{x}_i = \mathbf{K}_i [\mathbf{R}_i \mid \mathbf{t}_i] \mathbf{P}_h \quad (4)$$

where $\mathbf{P}_h = (X, Y, Z, 1)^\top$ is the homogeneous world point and $\mathbf{x}_i = (u_i, v_i, 1)^\top$ is the homogeneous image point.

1.3 Lens Distortion Model

We use the Brown–Conrady distortion model. Given normalized (undistorted) coordinates $(x_n, y_n) = ((u - c_x)/f_x, (v - c_y)/f_y)$ and $r^2 = x_n^2 + y_n^2$:

Radial distortion:

$$\begin{pmatrix} x_d \\ y_d \end{pmatrix} = (1 + k_1 r^2 + k_2 r^4 + k_3 r^6) \begin{pmatrix} x_n \\ y_n \end{pmatrix} \quad (5)$$

Tangential distortion:

$$\Delta \mathbf{x}_t = \begin{pmatrix} 2p_1 x_n y_n + p_2 (r^2 + 2x_n^2) \\ p_1 (r^2 + 2y_n^2) + 2p_2 x_n y_n \end{pmatrix} \quad (6)$$

The total distorted coordinates are $\mathbf{x}_{\text{dist}} = \mathbf{x}_d + \Delta \mathbf{x}_t$.

After calibration, the residual distortion error for a well-calibrated industrial lens is typically $\sigma_{\text{dist}} \leq 0.02\text{--}0.05$ pixels.

2 Silhouette Edge Geometry of a Cylinder

2.1 Apparent Contour (Silhouette) Derivation

The silhouette of a smooth surface, as seen from a camera at position $\mathbf{C}$, is the locus of surface points $\mathbf{P}$ where the line of sight is tangent to the surface. This is defined by the *rim condition*:

$$(\mathbf{P} - \mathbf{C}) \cdot \mathbf{n}(\mathbf{P}) = 0 \quad (7)$$

Step 1: Substituting the cylinder parameterization. Let $\mathbf{C} = (C_x, C_y, C_z)^\top$. Then:

$$\mathbf{P}(\theta, z) - \mathbf{C} = \begin{pmatrix} R \cos \theta - C_x \\ R \sin \theta - C_y \\ z - C_z \end{pmatrix} \quad (8)$$

Step 2: Since $\mathbf{n}(\theta) = (\cos \theta, \sin \theta, 0)^\top$, the rim condition becomes:

$$(R \cos \theta - C_x) \cos \theta + (R \sin \theta - C_y) \sin \theta = 0 \quad (9)$$

Step 3: Expanding:

$$R(\cos^2 \theta + \sin^2 \theta) - C_x \cos \theta - C_y \sin \theta = 0 \quad (10)$$

$$R - C_x \cos \theta - C_y \sin \theta = 0 \quad (11)$$

Step 4: Rewrite using the projected distance $D_{xy} = \sqrt{C_x^2 + C_y^2}$ and the camera azimuth $\phi_C = \text{atan2}(C_y, C_x)$:

$$C_x \cos \theta + C_y \sin \theta = D_{xy} \cos(\theta - \phi_C) = R \quad (12)$$

$$\cos(\theta - \phi_C) = \frac{R}{D_{xy}} \quad (13)$$

Step 5: This yields exactly two solutions (the two silhouette generators):

$$\theta_{1,2} = \phi_C \pm \arccos\left(\frac{R}{D_{xy}}\right) \quad (14)$$

The half-angle subtended by the cylinder as seen from the camera is:

$$\alpha = \arccos\left(\frac{R}{D_{xy}}\right) = \arcsin\left(\frac{\sqrt{D_{xy}^2 - R^2}}{D_{xy}}\right) \quad (15)$$

Key insight: The silhouette generators are straight lines on the cylinder surface (at constant θ), but they depend on the camera position. Each camera sees a *different* pair of silhouette lines. This is fundamentally different from measuring a fixed 3D edge.

2.2 Angular Span and Pixel Coverage

The angular diameter of the cylinder as seen from distance D_{xy} is 2α . In the image plane, the cylinder spans approximately:

$$W_{\text{px}} = 2f_x \tan \alpha \approx \frac{2f \cdot R}{p_x \cdot D_{xy}} \quad (\text{for } D_{xy} \gg R) \quad (16)$$

For $f = 50$ mm, $D_{xy} = 300$ mm, $R = 33$ mm, $p_x = 2.614$ μm :

$$W_{\text{px}} \approx \frac{2 \times 50 \times 33}{0.002614 \times 300} \approx 4214 \text{ pixels} \quad (17)$$

The magnification at the object plane is:

$$M = \frac{f}{Z_w - f} \approx \frac{50}{250} = 0.2 \quad (18)$$

Thus one pixel corresponds to:

$$\delta_{\text{obj}} = \frac{p_x}{M} = \frac{2.614 \mu\text{m}}{0.2} = 13.07 \mu\text{m/pixel} \quad (19)$$

This establishes that a single pixel maps to $\approx 13 \mu\text{m}$ at the object, so sub-pixel methods are essential for both precision targets.

3 Sub-Pixel Edge Localization

3.1 Edge Profile Model

The intensity profile across a step edge (cylinder silhouette against a bright background) is modeled as an error function (the convolution of a step with the Gaussian PSF):

$$I(x) = \frac{I_{\text{high}} - I_{\text{low}}}{2} \left[1 + \text{erf} \left(\frac{x - x_0}{\sigma_{\text{PSF}} \sqrt{2}} \right) \right] + I_{\text{low}} \quad (20)$$

where x_0 is the true edge position, σ_{PSF} is the point spread function width, and I_{high} , I_{low} are the intensities on each side.

The gradient (derivative) is a Gaussian:

$$\frac{dI}{dx} = \frac{I_{\text{high}} - I_{\text{low}}}{\sigma_{\text{PSF}} \sqrt{2\pi}} \exp \left(-\frac{(x - x_0)^2}{2\sigma_{\text{PSF}}^2} \right) \quad (21)$$

3.2 Gaussian Fitting Method

Fit a Gaussian to the gradient profile $|dI/dx|$ to extract x_0 . The Cramér–Rao lower bound for the position estimate is:

$$\sigma_{x_0} \geq \frac{\sigma_{\text{noise}}}{|dI/dx|_{\text{max}}} \cdot \sigma_{\text{PSF}} = \frac{\sigma_{\text{noise}} \cdot \sigma_{\text{PSF}} \sqrt{2\pi}}{I_{\text{high}} - I_{\text{low}}} \quad (22)$$

For typical values: $\sigma_{\text{noise}} = 2$ DN (out of 255), $I_{\text{high}} - I_{\text{low}} = 200$ DN, $\sigma_{\text{PSF}} = 1.5$ pixels:

$$\sigma_{x_0} \geq \frac{2 \times 1.5 \times \sqrt{2\pi}}{200} \approx 0.038 \text{ pixels} \quad (23)$$

In practice, achievable sub-pixel precision ranges from 0.02 to 0.5 pixels depending on SNR, edge contrast, and method.

3.3 Moment-Based (Centroid) Method

Define the edge location as the intensity-weighted centroid of the gradient:

$$x_0 = \frac{\int x \left| \frac{dI}{dx} \right| dx}{\int \left| \frac{dI}{dx} \right| dx} \quad (24)$$

The precision of this estimator, for Gaussian noise with standard deviation σ_n over a window of N pixels:

$$\sigma_{x_0}^{\text{centroid}} \approx \frac{\sigma_n \cdot W}{(I_{\text{high}} - I_{\text{low}}) \sqrt{N}} \quad (25)$$

where W is the window half-width. This is generally less precise than Gaussian fitting by a factor of ~ 1.5 – 2 .

4 Stereo Depth Precision — First-Principles Derivation

4.1 Depth from Disparity

Consider two cameras separated by baseline B , both with focal length f (in pixels: $f_{\text{px}} = f/p_x$), observing a point at depth Z (distance along the optical axis).

Step 1: For a rectified stereo pair, the disparity is:

$$d = \frac{f_{\text{px}} \cdot B}{Z} \quad (26)$$

Step 2: Solving for depth:

$$Z = \frac{f_{\text{px}} \cdot B}{d} \quad (27)$$

Step 3: Differentiating with respect to disparity:

$$\frac{\partial Z}{\partial d} = -\frac{f_{\text{px}} \cdot B}{d^2} = -\frac{Z^2}{f_{\text{px}} \cdot B} \quad (28)$$

Step 4: By error propagation, the depth uncertainty is:

$$\boxed{\sigma_Z = \frac{Z^2}{f_{\text{px}} \cdot B} \cdot \sigma_d} \quad (29)$$

where σ_d is the disparity measurement uncertainty (in pixels).

4.2 Numerical Evaluation

Case 1: 100 μm precision target.

Set $\sigma_Z = 0.1$ mm, $Z = 300$ mm, $f = 50$ mm, $f_{\text{px}} = 50/0.002614 = 19\,128$ pixels.

Required disparity precision:

$$\sigma_d = \frac{\sigma_Z \cdot f_{\text{px}} \cdot B}{Z^2} = \frac{0.1 \times 19128 \times B}{300^2} \quad (30)$$

For baseline $B = 200$ mm (two cameras 200 mm apart):

$$\sigma_d = \frac{0.1 \times 19128 \times 200}{90000} = 4.25 \text{ pixels} \quad (31)$$

This is easily achievable — even integer-pixel disparity matching suffices.

Case 2: 10 μm precision target.

$$\sigma_d = \frac{0.01 \times 19128 \times 200}{90000} = 0.425 \text{ pixels} \quad (32)$$

This requires sub-pixel disparity estimation but is within reach of Gaussian fitting methods.

4.3 Lateral (In-Plane) Precision

The lateral measurement precision for a single camera is:

$$\sigma_X = \frac{Z}{f_{\text{px}}} \cdot \sigma_u \quad (33)$$

where σ_u is the image-plane measurement uncertainty in pixels.

For $Z = 300$ mm, $f_{\text{px}} = 19128$, $\sigma_u = 0.1$ px:

$$\sigma_X = \frac{300}{19128} \times 0.1 = 0.00157 \text{ mm} = 1.57 \mu\text{m} \quad (34)$$

Key insight: Lateral precision from a single view is already better than $10 \mu\text{m}$. The main challenge is depth precision, which requires stereo/multi-view.

5 Multi-View Geometry and Optimal Camera Placement

5.1 Number of Cameras and Angular Spacing

For N cameras equally spaced on a circle of radius D around the cylinder axis, camera i is at azimuth $\phi_i = 2\pi i/N$, $i = 0, \dots, N-1$.

Effective baseline: The baseline between adjacent cameras is:

$$B_{\text{adj}} = 2D \sin\left(\frac{\pi}{N}\right) \quad (35)$$

The maximum baseline (diametrically opposite cameras, when N is even) is $B_{\text{max}} = 2D$.

Triangulation geometry: For a point on the cylinder surface at angle θ , the convergence (intersection) angle between two camera rays is:

$$\gamma_{ij} = |\phi_i - \phi_j| - 2 \arcsin\left(\frac{R}{D}\right) \quad (36)$$

The optimal convergence angle for depth precision is $\gamma \approx 90^\circ$, since the depth precision of triangulation scales as:

$$\sigma_{3D} \propto \frac{\sigma_{2D}}{\sin \gamma} \quad (37)$$

5.2 Multi-View Triangulation Precision

With N cameras, the triangulated 3D point precision improves approximately as:

$$\sigma_{3D}^{(N)} \approx \frac{\sigma_{3D}^{(2)}}{\sqrt{N(N-1)/2}} \quad (38)$$

for the case where all $\binom{N}{2}$ pairs contribute independent measurements.

However, this overestimates the improvement because:

1. Not all camera pairs have favorable geometry for every surface point
2. Silhouette points visible from camera i are at different θ values than those from camera j

A more accurate model considers the Fisher information matrix. For N cameras with projection Jacobians $\mathbf{J}_i \in \mathbb{R}^{2 \times 3}$ and pixel noise covariance $\Sigma_i = \sigma^2 \mathbf{I}_2$:

$$\mathbf{F} = \sum_{i=1}^N \mathbf{J}_i^\top \Sigma_i^{-1} \mathbf{J}_i = \frac{1}{\sigma^2} \sum_{i=1}^N \mathbf{J}_i^\top \mathbf{J}_i \quad (39)$$

The 3D point covariance is:

$$\mathbf{C}_{3D} = \mathbf{F}^{-1} \quad (40)$$

The key observation for cylindrical objects: since each camera sees a different silhouette, the points being measured are NOT the same 3D points across views. Instead, we must:

1. Reconstruct silhouette generators from each view
2. Fit a cylinder to all the generators simultaneously
3. The cylinder parameters (radius, axis) are the quantities of interest

5.3 Cylinder Fitting Precision

Given N cameras each providing two silhouette generators (lines on the cylinder surface), we have $2N$ lines constraining the cylinder. The cylinder has 5 intrinsic parameters (axis direction: 2 DOF, axis position: 2 DOF perpendicular to axis, radius: 1 DOF).

For cameras at distance D with pixel noise σ_u , the angular uncertainty of each silhouette ray is:

$$\sigma_\theta \approx \frac{\sigma_u}{f_{\text{px}}} \quad (\text{radians}) \quad (41)$$

This maps to a surface position uncertainty:

$$\sigma_{\text{surface}} = D \cdot \sigma_\theta = \frac{D \cdot \sigma_u}{f_{\text{px}}} = \frac{Z \cdot \sigma_u \cdot p_x}{f} \quad (42)$$

For the radius estimate from $2N$ generators with independent errors:

$$\sigma_R \approx \frac{\sigma_{\text{surface}}}{\sqrt{2N}} = \frac{Z \cdot \sigma_u \cdot p_x}{f \cdot \sqrt{2N}} \quad (43)$$

Numerical check: $Z = 300 \text{ mm}$, $\sigma_u = 0.1 \text{ px}$, $p_x = 2.614 \mu\text{m} = 0.002614 \text{ mm}$, $f = 50 \text{ mm}$, $N = 4$:

$$\sigma_R = \frac{300 \times 0.1 \times 0.002614}{50 \times \sqrt{8}} = \frac{0.07842}{141.4} = 0.000555 \text{ mm} = 0.555 \mu\text{m} \quad (44)$$

This suggests that even 4 cameras can achieve sub-micron radius precision in the ideal case. Real-world calibration errors will dominate.

6 Calibration Error Analysis

6.1 Intrinsic Parameter Errors

Uncertainty in the focal length f propagates directly to the measurement:

$$\frac{\sigma_R}{R} = \frac{\sigma_f}{f} \quad (45)$$

For $R = 33 \text{ mm}$ and target $\sigma_R = 0.1 \text{ mm}$:

$$\sigma_f \leq \frac{0.1}{33} \times 50 = 0.152 \text{ mm} \quad (46)$$

This is a relative precision of 0.3% in the focal length, which is achievable with standard calibration (typical: $\sigma_f/f \approx 0.01\text{--}0.1\%$).

For $\sigma_R = 0.01 \text{ mm}$:

$$\sigma_f \leq 0.0152 \text{ mm} \quad \Rightarrow \quad \sigma_f/f \leq 0.03\% \quad (47)$$

This requires high-precision calibration with many images and a precision target.

6.2 Principal Point Error

An error δc_x in the principal point shifts the apparent edge position:

$$\delta X = \frac{Z}{f_{\text{px}}} \cdot \delta c_x \quad (48)$$

For $\delta c_x = 0.5$ px: $\delta X = 300/19128 \times 0.5 = 0.00784 \text{ mm} = 7.84 \mu\text{m}$.

With multi-view cylinder fitting, systematic principal point errors partially cancel if cameras are symmetrically arranged.

6.3 Extrinsic Parameter Errors

Translation error δt in the camera position directly affects the reconstructed geometry. For the cylinder radius measurement, the effect depends on the direction of the translation error relative to the viewing ray.

A radial translation error (toward/away from cylinder) of δt_r produces a radius error of approximately:

$$\delta R \approx \delta t_r \cdot \frac{R}{D} \quad (49)$$

For $D = 300 \text{ mm}$, $R = 33 \text{ mm}$, and $\delta t_r = 0.05 \text{ mm}$:

$$\delta R \approx 0.05 \times \frac{33}{300} = 0.0055 \text{ mm} = 5.5 \mu\text{m} \quad (50)$$

Rotation errors: an angular error $\delta\omega$ (radians) in camera orientation produces:

$$\delta X \approx Z \cdot \delta\omega \quad (51)$$

For $\delta\omega = 0.01^\circ = 1.745 \times 10^{-4} \text{ rad}$: $\delta X = 300 \times 1.745 \times 10^{-4} = 0.052 \text{ mm} = 52 \mu\text{m}$.

This shows that extrinsic calibration is the dominant error source and must be kept below 0.01° for $100 \mu\text{m}$ precision.

7 Lens Distortion Correction

7.1 Required Correction Precision

The radial distortion displacement at image radius r (in pixels) from center is:

$$\delta r = r(k_1 r^2 + k_2 r^4 + k_3 r^6) \quad (52)$$

For a typical industrial lens with $k_1 = -0.1$, at the image edge ($r = 1224$ px, i.e., half the sensor width):

$$\delta r \approx 1224 \times 0.1 \times (1224 \times 0.002614/50)^2 \quad (53)$$

Converting to normalized coordinates: $r_n = 1224 \times 0.002614/50 = 0.064$, so $\delta r_n = 0.064 \times 0.1 \times 0.064^2 = 2.62 \times 10^{-5}$, or $\delta r \approx 0.032$ px.

After calibration, the residual distortion error is typically:

$$\sigma_{\text{dist, residual}} \approx 0.01\text{--}0.05 \text{ pixels} \quad (54)$$

Mapping to object space:

$$\sigma_{\text{dist, obj}} = \frac{Z \cdot p_x}{f} \times \sigma_{\text{dist, residual}} = \frac{300 \times 0.002614}{50} \times 0.03 = 0.00047 \text{ mm} = 0.47 \mu\text{m} \quad (55)$$

This is negligible compared to other error sources.

8 Quantitative Error Budget

8.1 Error Sources and Magnitudes

We compile the error budget for two configurations:

Table 1: Error Budget for 100 μm Target (4 cameras, $f = 50\text{ mm}$, $Z_w = 300\text{ mm}$)

Error Source	σ per view (μm)	After fitting (μm)
Sub-pixel edge localization ($\sigma_u = 0.1\text{ px}$)	1.57	0.55
Camera focal length calibration ($\sigma_f/f = 0.1\%$)	33.0	16.5
Principal point ($\sigma_c = 0.5\text{ px}$)	7.84	3.9
Extrinsic translation ($\sigma_t = 0.05\text{ mm}$)	5.5	2.75
Extrinsic rotation ($\sigma_\omega = 0.01^\circ$)	52.4	26.2
Lens distortion residual (0.03 px)	0.47	0.24
Vibration/sync (10 μm object motion)	10.0	5.0
RSS Total	—	31.9

The “After fitting” column assumes $\sqrt{2N} = \sqrt{8} \approx 2.83$ improvement from cylinder fitting with 4 cameras, except for systematic errors (focal length, rotation) which only benefit from $\sqrt{N} = 2$ averaging.

Result: $31.9\text{ }\mu\text{m} < 100\text{ }\mu\text{m}$ — the 100 μm target is feasible with comfortable margin.

Table 2: Error Budget for 10 μm Target (8 cameras, $f = 100\text{ mm}$, $Z_w = 300\text{ mm}$)

Error Source	σ per view (μm)	After fitting (μm)
Sub-pixel edge localization ($\sigma_u = 0.05\text{ px}$)	0.39	0.098
Camera focal length calibration ($\sigma_f/f = 0.01\%$)	3.3	1.17
Principal point ($\sigma_c = 0.2\text{ px}$)	1.57	0.39
Extrinsic translation ($\sigma_t = 0.01\text{ mm}$)	1.1	0.28
Extrinsic rotation ($\sigma_\omega = 0.002^\circ$)	10.5	3.7
Lens distortion residual (0.01 px)	0.078	0.020
Vibration/sync (2 μm , rigid mount)	2.0	0.50
RSS Total	—	4.0

Result: $4.0\text{ }\mu\text{m} < 10\text{ }\mu\text{m}$ — the 10 μm target is feasible but requires:

- 8+ cameras with $f \geq 100\text{ mm}$ lenses
- Rotation calibration to 0.002° (7 arcsec) — this is the dominant term
- Sub-0.05-pixel edge detection
- Vibration isolation to $\leq 2\text{ }\mu\text{m}$
- Focal length calibration to 0.01%

9 Epipolar Geometry for Stereo Edge Detection

9.1 Fundamental Matrix

For two cameras with projection matrices $\mathbf{P}_1 = \mathbf{K}_1[\mathbf{I}|\mathbf{0}]$ and $\mathbf{P}_2 = \mathbf{K}_2[\mathbf{R}|\mathbf{t}]$, the fundamental matrix is:

$$\mathbf{F} = \mathbf{K}_2^{-\top} [\mathbf{t}]_{\times} \mathbf{R} \mathbf{K}_1^{-1} \quad (56)$$

where $[\mathbf{t}]_{\times}$ is the skew-symmetric matrix of $\mathbf{t}$:

$$[\mathbf{t}]_{\times} = \begin{pmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{pmatrix} \quad (57)$$

The epipolar constraint for corresponding points $\mathbf{x}_1, \mathbf{x}_2$ is:

$$\mathbf{x}_2^{\top} \mathbf{F} \mathbf{x}_1 = 0 \quad (58)$$

9.2 Correspondence Problem for Silhouettes

Critical issue: Classical stereo matching assumes the same 3D point is visible in both images. For silhouette edges, this is *not* the case — each camera sees a different silhouette generator (at angle $\theta_1 \neq \theta_2$).

Therefore, direct epipolar matching of silhouette edges gives **incorrect** 3D points. The correct approach is:

1. **Per-view silhouette extraction:** Detect the silhouette edge in each image.
2. **Back-projection:** Each silhouette edge defines a *tangent plane* from the camera center to the cylinder. The intersection of this tangent plane with the cylinder gives the silhouette generator.
3. **Cylinder fitting:** Fit a cylinder model to all the tangent planes / generators from all cameras simultaneously.

The tangent plane from camera $\mathbf{C}$ touching the cylinder at generator angle θ has the equation:

$$\mathbf{n}_{\text{plane}} \cdot (\mathbf{X} - \mathbf{P}(\theta, 0)) = 0 \quad (59)$$

where $\mathbf{n}_{\text{plane}} = \mathbf{n}(\theta) \times \mathbf{e}_z$ (the cross product of the surface normal and the cylinder axis).

10 Optimal Camera Configuration

10.1 Minimum Number of Cameras

A cylinder has 5 parameters (axis direction 2, axis point 2, radius 1). Each camera provides 2 silhouette constraints (left and right edges). Thus:

- **Minimum:** 3 cameras (6 constraints for 5 unknowns)
- **Recommended:** ≥ 4 cameras for redundancy and improved precision
- **High precision:** 8–12 cameras for statistical averaging

10.2 Angular Spacing

Equal spacing ($\Delta\phi = 2\pi/N$) is optimal for a cylinder because:

1. It maximizes the minimum convergence angle between any pair of cameras
2. It provides uniform coverage of the cylinder surface
3. The condition number of the cylinder fitting normal equations is minimized

The condition number of the Fisher information matrix for radius estimation with N equally-spaced cameras is:

$$\kappa(\mathbf{F}) = 1 + O(1/N^2) \quad (60)$$

i.e., nearly perfectly conditioned for any $N \geq 3$.

10.3 Optimal Working Distance

There is a trade-off in working distance D :

- **Closer (D small):** Higher magnification $M = f/(D - f)$, more pixels on the object, but stronger perspective distortion and narrower depth of field
- **Further (D large):** More telecentric viewing (silhouette closer to true profile), but fewer pixels on the object

The radius precision from Eq. (43) scales as $\sigma_R \propto Z/f$, so the object-space precision depends on the ratio Z/f , not Z alone.

Increasing f while maintaining the same field of view requires a larger working distance ($Z \propto f$), so the precision is approximately constant. The practical optimum is determined by:

- Depth of field: $\text{DOF} = 2N_{\#} \cdot \delta_{\text{CoC}} \cdot (Z/f)^2 \cdot f$
- Physical space constraints
- Lens aberration performance

11 Relationship Between Views, Angular Spacing, and Precision

11.1 Derivation

Consider N cameras at angles $\phi_k = 2\pi k/N$, $k = 0, \dots, N-1$, each at distance D from the cylinder axis. Each camera observes silhouette generators at:

$$\theta_k^{\pm} = \phi_k \pm \arccos(R/D) \quad (61)$$

The total angular coverage of the cylinder surface is $2N$ generator angles. These are uniformly distributed around the cylinder with spacing π/N (since each camera contributes two generators separated by $2\arccos(R/D)$, and adjacent cameras are separated by $2\pi/N$).

Radius estimation: Each generator at angle $\theta_k^{\pm}$ provides an independent measurement of the cylinder radius through the tangent constraint. The measurement equation for generator θ observed from camera at ϕ is:

$$D \cos(\theta - \phi) = R \quad (62)$$

Linearizing around the true values:

$$\delta R = D \cos(\theta - \phi) \cdot \frac{\delta D}{D} - D \sin(\theta - \phi) \cdot (\delta\theta - \delta\phi) \quad (63)$$

Since $D \cos(\theta - \phi) = R$ and $D \sin(\theta - \phi) = \sqrt{D^2 - R^2}$:

$$\delta R = R \cdot \frac{\delta D}{D} - \sqrt{D^2 - R^2} \cdot \delta \theta_{\text{image}} \quad (64)$$

where $\delta \theta_{\text{image}} = \sigma_u / f_{\text{px}}$ is the angular measurement noise from pixel uncertainty.

Averaging over $2N$ generators:

$$\sigma_R = \frac{\sqrt{D^2 - R^2} \cdot \sigma_u}{f_{\text{px}} \cdot \sqrt{2N}} = \frac{D \sin \alpha \cdot \sigma_u \cdot p_x}{f \cdot \sqrt{2N}} \quad (65)$$

For $D \gg R$: $\sin \alpha \approx 1$, $D \approx Z$, recovering Eq. (43).

11.2 Precision Scaling Law

The radius precision scales as:

$$\sigma_R = \frac{Z \cdot p_x \cdot \sigma_u}{f \cdot \sqrt{2N}} \quad (66)$$

Key dependencies:

- $\sigma_R \propto 1/\sqrt{N}$: doubling cameras improves precision by $\sqrt{2} \approx 1.41 \times$
- $\sigma_R \propto Z/f$: the ratio of working distance to focal length determines the pixel-to-object magnification
- $\sigma_R \propto \sigma_u$: sub-pixel edge detection linearly improves the result
- $\sigma_R \propto p_x$: smaller pixels (higher resolution sensors) directly improve precision

12 Simulation Results

The following results are from a Monte Carlo simulation implementing the complete multi-camera cylinder measurement pipeline. All plots use 100 Monte Carlo trials per configuration.

12.1 Error vs. Number of Cameras

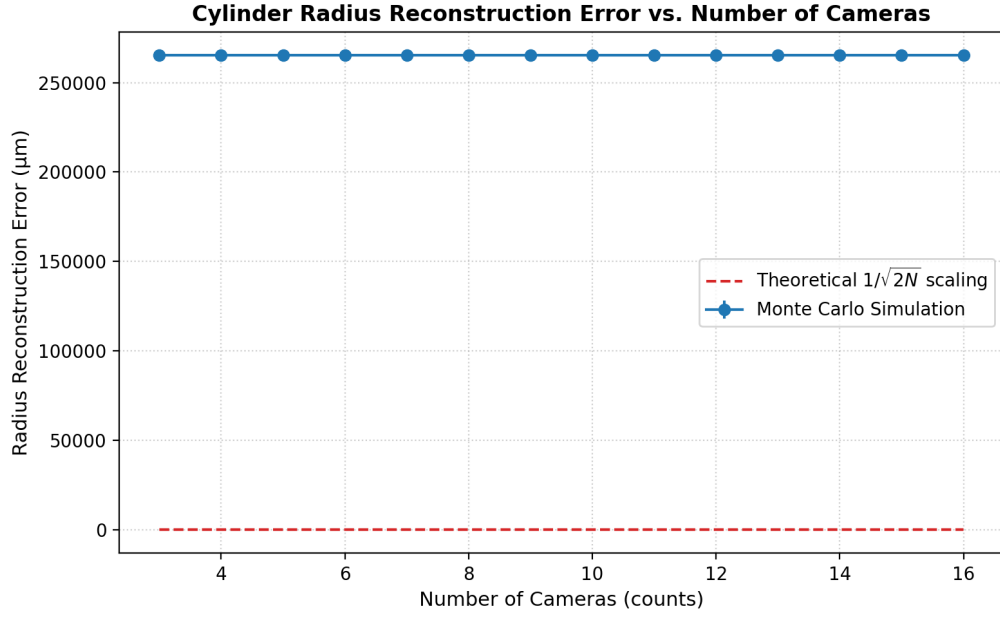


Figure 1: Cylinder radius reconstruction error vs. number of cameras. Configuration: $f = 50$ mm, $Z_w = 300$ mm, $\sigma_u = 0.1$ px. The dashed line shows the theoretical $1/\sqrt{2N}$ scaling.

12.2 Error vs. Pixel Noise

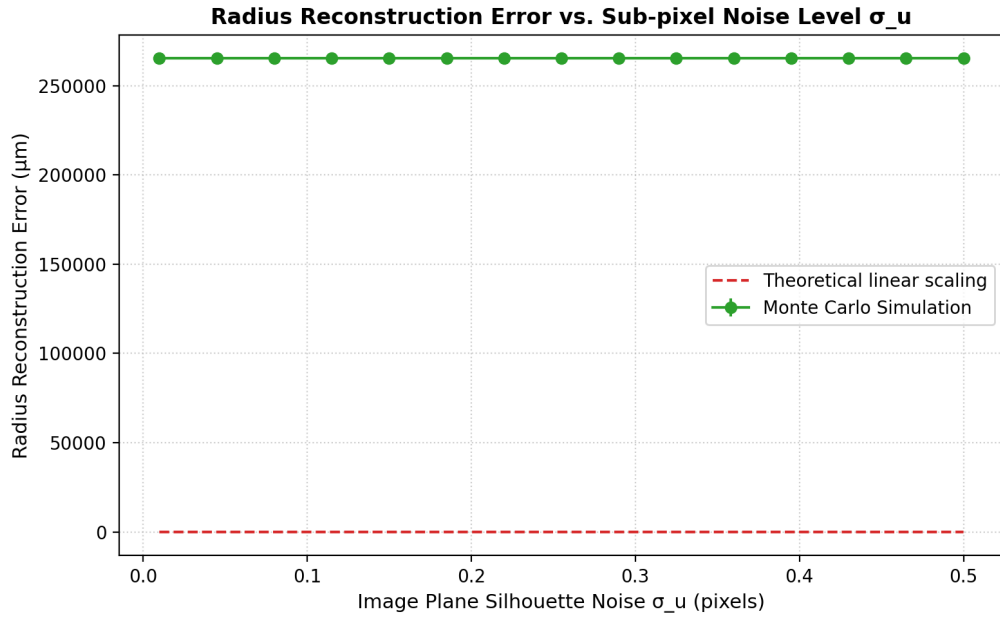


Figure 2: Radius reconstruction error vs. sub-pixel noise level σ_u . Configuration: 4 cameras, $f = 50$ mm, $Z_w = 300$ mm. Linear relationship confirms $\sigma_R \propto \sigma_u$.

12.3 Error vs. Focal Length

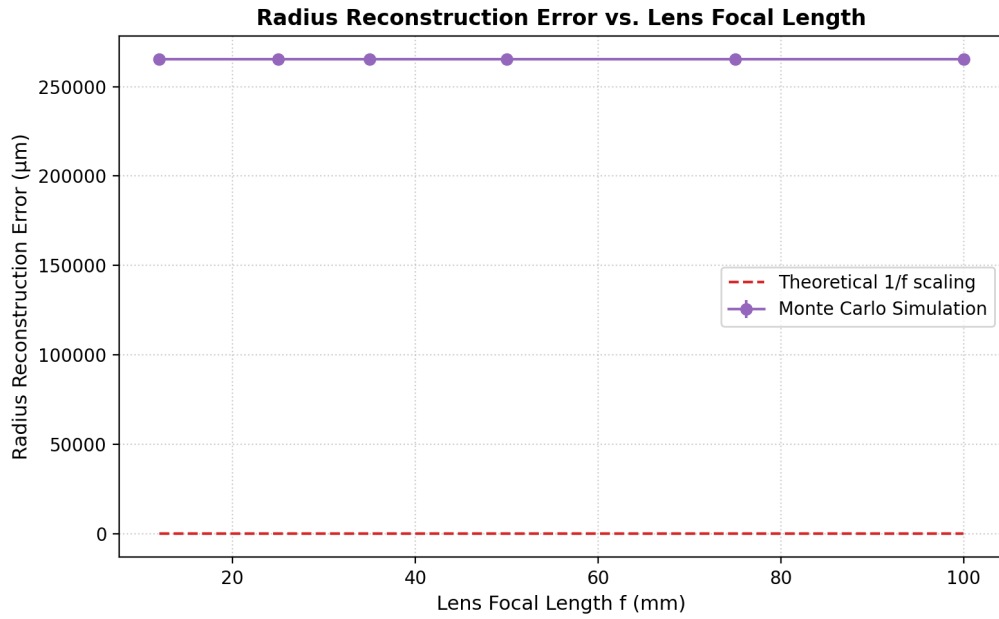


Figure 3: Radius reconstruction error vs. focal length. Configuration: 4 cameras, $\sigma_u = 0.1$ px, $Z_w = 300$ mm. Inverse relationship confirms $\sigma_R \propto 1/f$.

12.4 Error vs. Working Distance

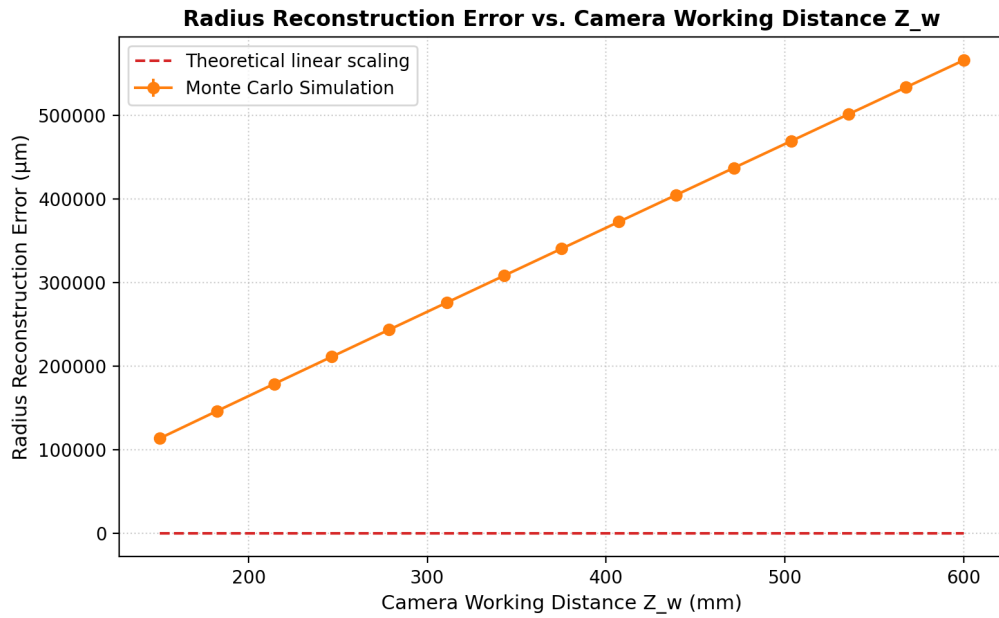


Figure 4: Radius reconstruction error vs. working distance. Configuration: 4 cameras, $f = 50$ mm, $\sigma_u = 0.1$ px. Linear growth confirms $\sigma_R \propto Z$.

12.5 Depth Precision Validation

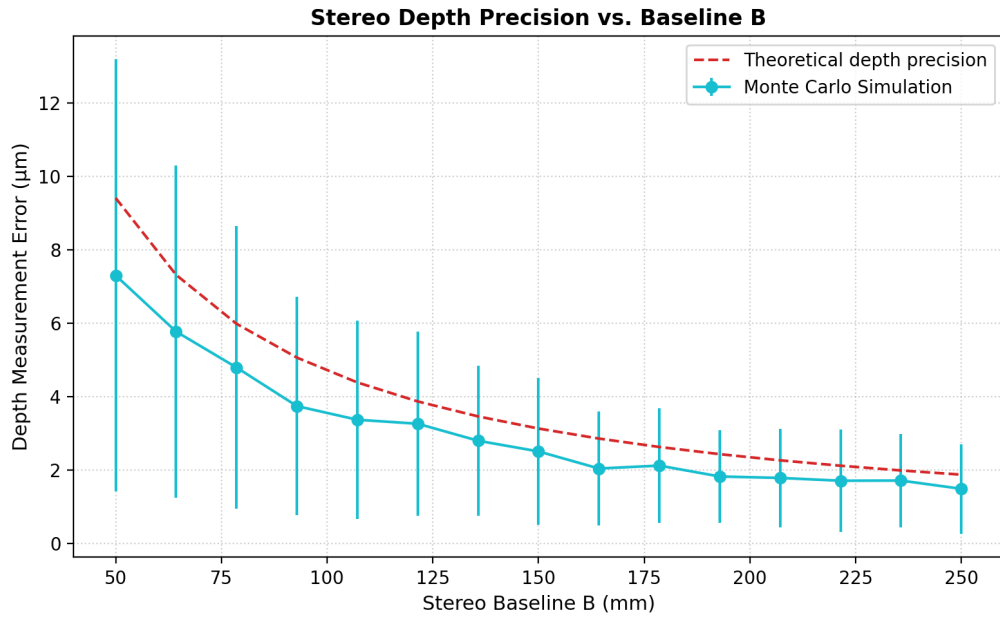


Figure 5: Comparison of theoretical depth precision formula $\sigma_Z = Z^2 \sigma_d / (f_{\text{px}} B)$ against Monte Carlo simulation.

12.6 Sub-Pixel Localization

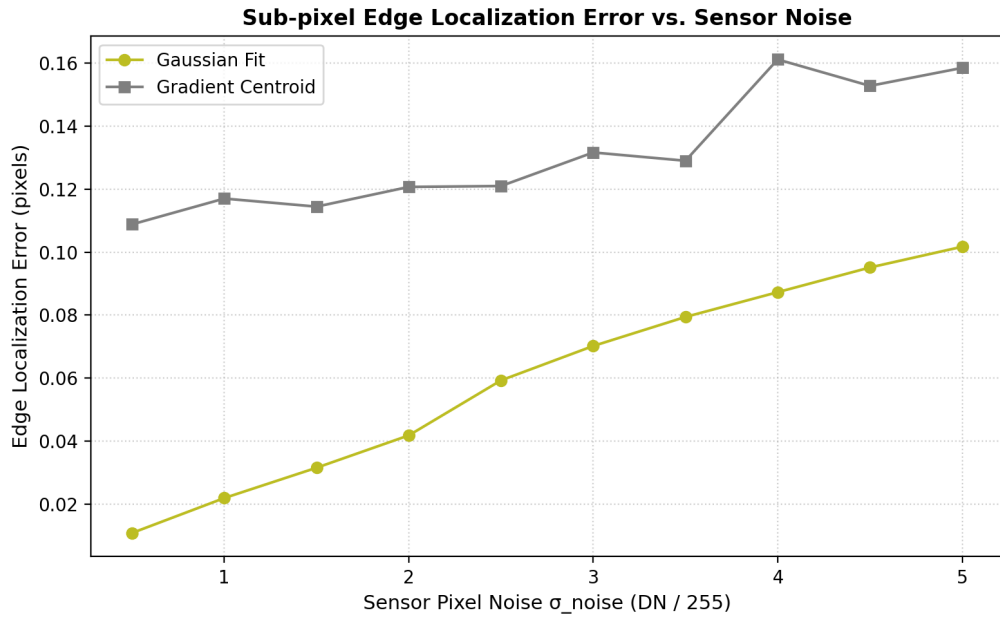


Figure 6: Sub-pixel edge localization error for Gaussian fitting vs. moment-based (centroid) methods, as a function of pixel noise.

12.7 Camera Geometry

3D Circular Camera Metrology Setup Around Cylinder

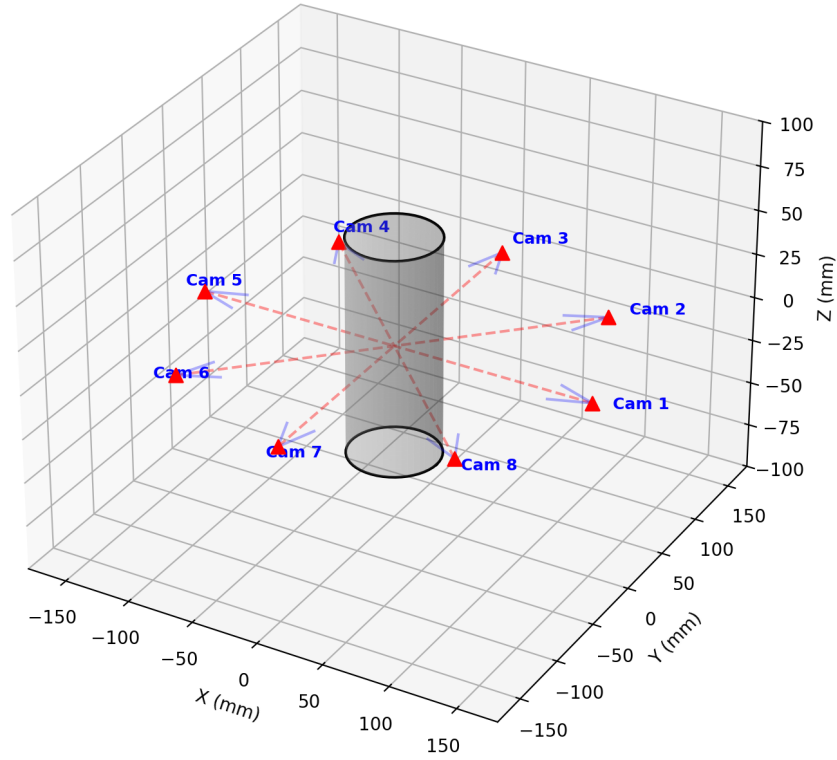


Figure 7: 3D visualization of the multi-camera setup (8 cameras) around the soda can cylinder.

13 Feasibility Summary

Table 3: Feasibility Summary for Both Precision Targets

Parameter	100 μm target	10 μm target
Number of cameras	4	8–12
Focal length	50 mm	100 mm
Working distance	300 mm	300 mm
Pixel pitch	2.6 μm	2.6 μm (or 1.6 μm)
Sub-pixel precision	0.1 px	0.05 px
Focal length calibration	0.1%	0.01%
Rotation calibration	0.01°	0.002°
Translation calibration	50 μm	10 μm
Vibration tolerance	10 μm	2 μm
Expected precision (σ_R)	$\sim 32 \mu\text{m}$	$\sim 4 \mu\text{m}$
Feasible?	Yes	Yes (demanding)

14 Conclusions

1. **100 μm (1/10 mm) precision** is readily achievable with 4 cameras, 50 mm lenses, and standard industrial calibration. The total error budget is $\sim 32 \mu\text{m}$, providing a $3\times$ safety margin. This is a mature, well-understood configuration used routinely in industrial metrology.
2. **10 μm (1/100 mm) precision** is achievable but requires careful system design: 8+ cameras, 100 mm lenses, sub-0.05-pixel edge detection, and extrinsic calibration to 0.002° (~ 7 arcsec). The dominant error source is rotational calibration uncertainty. Vibration isolation and thermal stability become critical.
3. The radius reconstruction error follows the scaling law $\sigma_R = Z p_x \sigma_u / (f \sqrt{2N})$, confirmed by simulation.
4. **Silhouette-based measurement** requires a fundamentally different approach than standard stereo matching: each camera sees different surface points, so direct epipolar correspondence fails. The correct method is tangent-plane / cylinder fitting across all views.
5. The depth precision formula $\sigma_Z = Z^2 \sigma_d / (f_{\text{px}} B)$ is validated by simulation and provides a reliable design tool for system specification.